

Solution Code - JUnit, Mockito and SL4J - JUnit_Basic_Assertions

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: src\test\java\com\example\AppTest.java

```
1 | package com.example;
2 |
3 | import org.junit.Test;
4 | import static org.junit.Assert.*;
5 |
6 | public class AssertionsTest {
7 |     @Test
8 |     public void testAssertions() {
9 |         assertEquals(5, 2 + 3);
10 |        assertTrue(5 > 3);
11 |        assertFalse(5 < 3);
12 |        assertNull(null);
13 |        assertNotNull(new Object());
14 |    }
15 | }
```

Source File: src\test\java\com\example\BasicAssertionsTest.java

```
1 | package com.example;
2 |
3 | import org.junit.Test;
4 | import static org.junit.Assert.*;
5 |
6 | public class BasicAssertionsTest {
7 |     @Test
8 |     public void testBasicMath() {
9 |         int expected = 10;
10 |        int actual = 5 + 5;
11 |        assertEquals("Basic addition failed", expected, actual);
12 |    }
13 |
14 |    @Test
15 |    public void testArrayEquality() {
16 |        int[] expected = {1, 2, 3};
17 |        int[] actual = {1, 2, 3};
18 |        assertEquals("Arrays are not equal", expected, actual);
19 |    }
20 |
21 |    @Test
22 |    public void testNullsAndBoolean() {
23 |        Object obj = null;
24 |        assertNull("Object should be null", obj);
25 |        assertTrue("Condition should be true", 5 > 3);
26 |    }
27 | }
```

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```
26 |         assertFalse("Condition should be false", 3 > 5);
27 |     }
28 | }
```